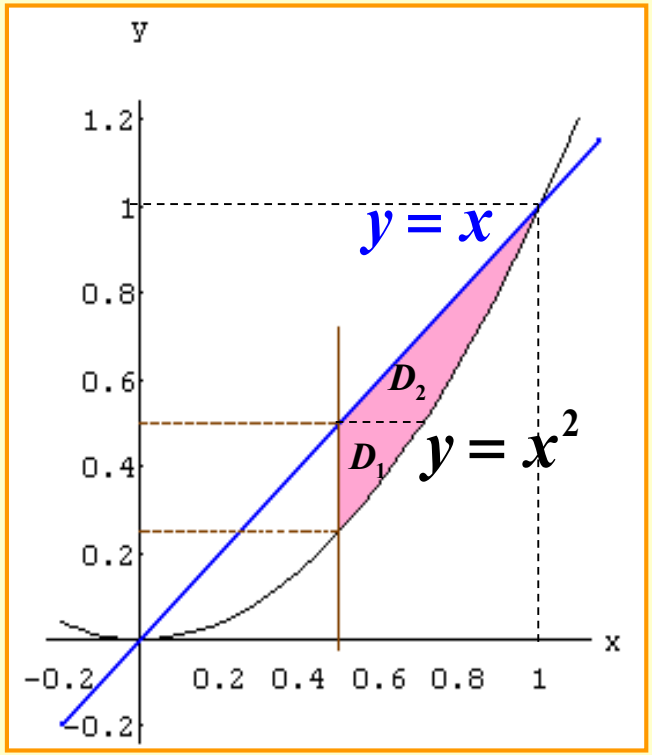
例　计算积分 $I=\int_{\frac{1}{4}}^{\frac{1}{2}}dy\int_{\frac{1}{2}}^{\sqrt{y}}e^{\frac{y}{x}}dx+\int_{\frac{1}{2}}^{1}dy\int_{y}^{\sqrt{y}}e^{\frac{y}{x}}dx.$

解　$\because\int e^{\frac{y}{x}}dx$不能用初等函数表示　$\therefore$先改变积分次序.

$$D_1:\begin{cases}\dfrac{1}{2}\le x\le\sqrt{y}\\ \dfrac{1}{4}\le y\le\dfrac{1}{2}\end{cases}\qquad D_2:\begin{cases}y\le x\le\sqrt{y}\\ \dfrac{1}{2}\le y\le 2\end{cases}$$

$$D=D_1+D_2:\begin{cases}x^2\le y\le x\\ \dfrac{1}{2}\le x\le 1\end{cases}$$



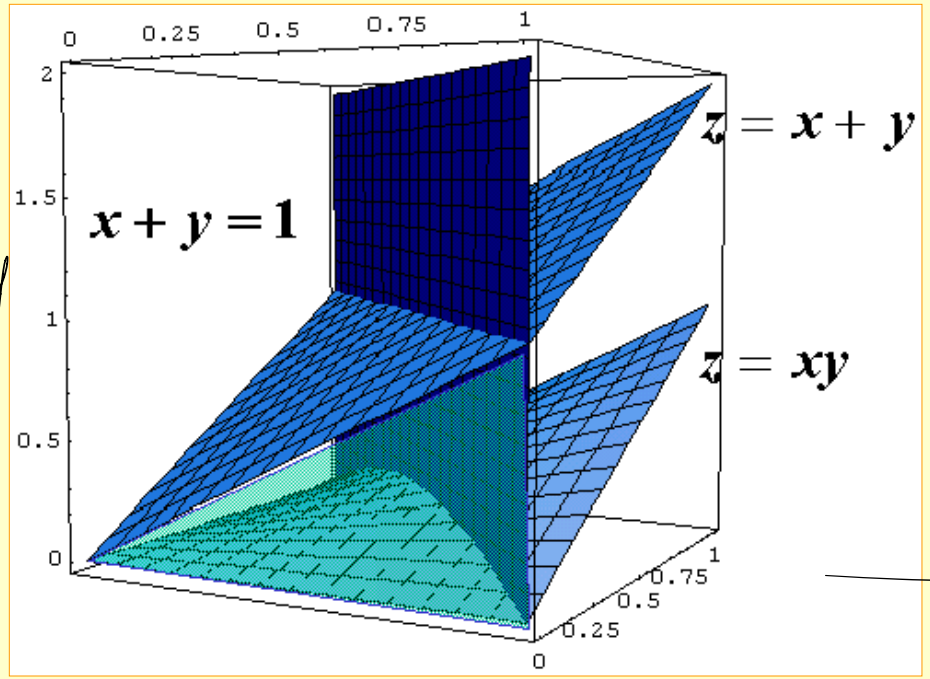

$$I=\iint_{D_1+D_2}e^{\frac{y}{x}}dxdy=\int_{\frac{1}{2}}^{1}dx\int_{x^2}^{x}e^{\frac{y}{x}}dy$$

$$=\int_{\frac{1}{2}}^{1}x(e-e^x)dx=\frac{3}{8}e-\frac{1}{2}\sqrt{e}.$$

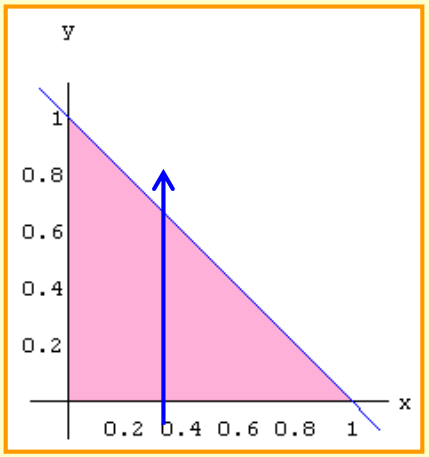
**例** 求由下列曲面所围成的立体体积，

$z = x + y$，$z = xy$，$x + y = 1$，$x = 0$，$y = 0$.

**解** 曲面围成的立体如图.



所围立体在$xoy$面上的投影是




$$\because \quad 0 \le x+y \le 1, \ \therefore \ x+y \ge xy,$$

$$\text{所求体积} V = \iint_D (x+y-xy)\,d\sigma$$

$$= \int_0^1 dx \int_0^{1-x} (x+y-xy)\,dy$$

$$= \int_0^1 [x(1-x) + \frac{1}{2}(1-x)^3]\,dx = \frac{7}{24}.$$

小结二：

二重积分在直角坐标下的计算公式

$$\iint_D f(x,y)d\sigma = \int_a^b dx \int_{\varphi_1(x)}^{\varphi_2(x)} f(x,y)dy. \qquad [\text{X—型}]$$

$$\iint_D f(x,y)d\sigma = \int_c^d dy \int_{\varphi_1(y)}^{\varphi_2(y)} f(x,y)dx. \qquad [\text{Y—型}]$$

（在积分中要正确选择**积分次序**）

口诀：后积先定限，中间划根线，

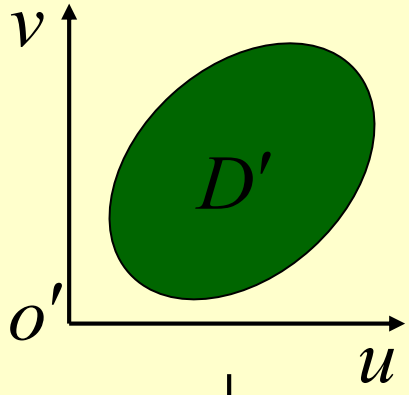
先交是下限，后交是上限。

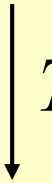
## 三、二重积分的换元法

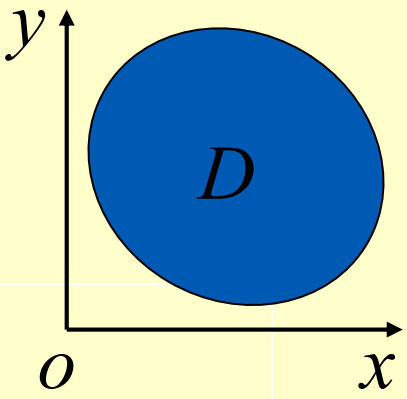
不要求掌握

**定理:** 设 $f(x,y)$ 在闭域 $D$ 上连续变换:

$$T:\begin{cases} x = x(u,v) \\ y = y(u,v) \end{cases} (u,v) \in D' \to D$$




满足 (1) $x(u,v), y(u,v)$ 在 $D'$ 上 一阶导数连续;

(2) 在 $D'$ 上 雅可比行列式

$$J(u,v) = \frac{\partial(x,y)}{\partial(u,v)} \neq 0$$

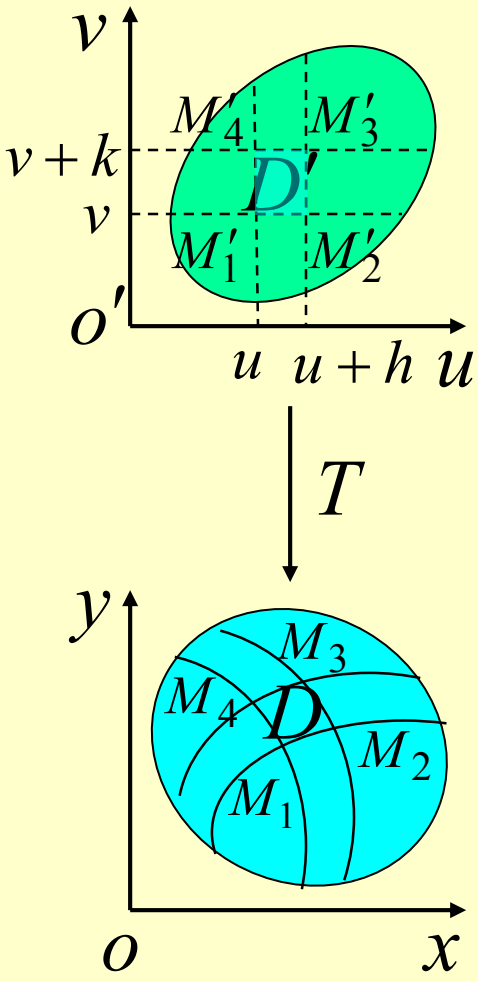
(3) 变换 $T: D' \to D$ 是一一对应的 ,

则 $$\iint_D f(x,y)\,\mathrm{d}x\,\mathrm{d}y = \iint_{D'} f(x(u,v),y(u,v))\,|J(u,v)|\,\mathrm{d}u\,\mathrm{d}v$$

*证: 根据定理条件可知变换 $T$ 可逆.

在$uo'v$坐标面上, 用平行于坐标轴的直线分割区域$D'$, 任取其中一个小矩形, 其顶点为

$$M_1'(u,v), \quad M_2'(u+h,v),$$
$$M_3'(u+h,v+k), \quad M_4'(u,v+k).$$




通过变换$T$, 在 $xoy$ 面上得到一个四边形, 其对应顶点为$M_i(x_i,y_i)\ (i=1,2,3,4)$

令 $\rho=\sqrt{h^2+k^2}$, 则

$$x_2-x_1=x(u+h,v)-x(u,v)=\left.\frac{\partial x}{\partial u}\right|_{(u,v)}h+o(\rho)$$

$$x_4 - x_1 = x(u, v+k) - x(u, v = \left.\frac{\partial x}{\partial v}\right|_{(u,v)} k + o(\rho)$$

同理得 $y_2 - y_1 = \left.\frac{\partial y}{\partial u}\right|_{(u,v)} h + o(\rho)$

$$y_4 - y_1 = \left.\frac{\partial y}{\partial v}\right|_{(u,v)} k + o(\rho)$$

当$h, k$充分小时, 曲边四边形$M_1M_2M_3M_4$近似于平行四边形, 故其面积近似为

$$\Delta\sigma \approx \left| \overrightarrow{M_1M_2} \times \overrightarrow{M_1M_4} \right| = \left| \begin{vmatrix} x_2 - x_1 & y_2 - y_1 \\ x_4 - x_1 & y_4 - y_1 \end{vmatrix} \right|$$

$$\approx \left| \begin{vmatrix} \frac{\partial x}{\partial u} h & \frac{\partial y}{\partial u} k \\ \frac{\partial x}{\partial v} h & \frac{\partial y}{\partial v} k \end{vmatrix} \right| = \left| \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} \right| hk = \left| J(u,v) \right| hk$$

因此面积元素的关系为 $\mathrm{d}\sigma = |J(u,v)|\,\mathrm{d}u\,\mathrm{d}v$

从而得二重积分的换元公式:

$$\iint_D f(x,y)\,\mathrm{d}x\,\mathrm{d}y$$

$$= \iint_{D'} f(x(u,v),y(u,v))\ |J(u,v)|\,\mathrm{d}u\,\mathrm{d}v$$

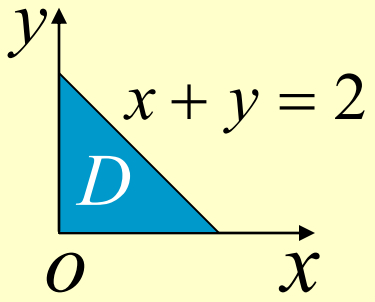
**例如,** 直角坐标转化为极坐标时, $x = r\cos\theta$, $y = r\sin\theta$

$$J = \frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & -r\cos\theta \end{vmatrix} = r$$

$$\therefore \iint_D f(x,y)\,\mathrm{d}x\,\mathrm{d}y$$

$$= \iint_{D'} f(r\cos\theta, r\sin\theta)\,r\,\mathrm{d}r\,\mathrm{d}\theta$$

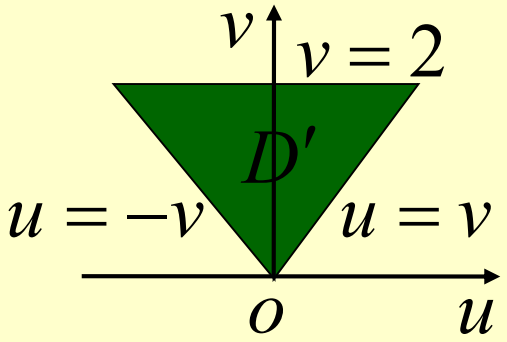
**例1.** 计算 $\iint_D e^{\frac{y-x}{y+x}}\,\mathrm{d}x\,\mathrm{d}y$, 其中$D$ 是 $x$ 轴 $y$ 轴和直线 $x+y=2$ 所围成的闭域.




**解:** 令 $u=y-x$, $v=y+x$,则

$$x=\frac{v-u}{2},\ y=\frac{v+u}{2}\qquad (D'\to D)$$

$$J=\frac{\partial(x,y)}{\partial(u,v)}=\begin{vmatrix}-\frac{1}{2} & \frac{1}{2}\\ \frac{1}{2} & \frac{1}{2}\end{vmatrix}=\frac{-1}{2}$$




$$\therefore \iint_D e^{\frac{y-x}{y+x}}\,\mathrm{d}x\,\mathrm{d}y=\iint_{D'} e^{\frac{u}{v}}\left|\frac{-1}{2}\right|\mathrm{d}u\,\mathrm{d}v=\frac{1}{2}\int_0^2\mathrm{d}v\int_{-v}^{v}e^{\frac{u}{v}}\,\mathrm{d}u$$

$$=\frac{1}{2}\int_0^2(e-e^{-1})v\,\mathrm{d}v=e-e^{-1}$$

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二重积分化成极坐标系下的二重积分

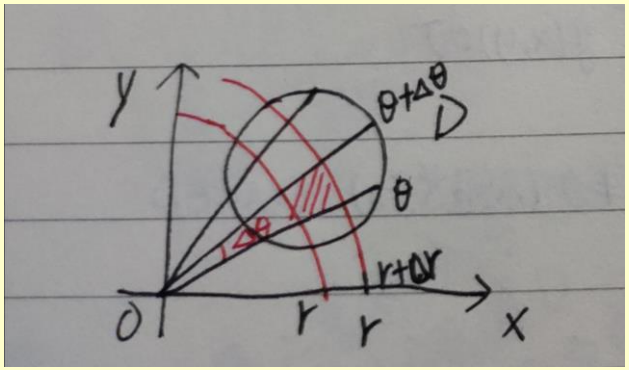
若 $\iint_D f(x,y)\,d\sigma$ 存在，当 $f(x,y)$ 中含有 $x^2+y^2$（$\sqrt{x^2+y^2}$）or 积分区域 $D$ 是圆域 or 圆周与直线围成的区域

令 $x=r\cos\theta$　$y=r\cos\theta$　则 $\iint_D f(x,y)\,d\sigma = \iint_{D_{r\theta}} f(r\cos\theta, r\sin\theta)\,r\,d\theta\,dr$

计算定积分　$Q=\int_a^b f(x)\,dx \Leftrightarrow dQ = f(x)dx \quad x\in[a,b]$

设 $Q=\iint_D f(x,y)\,d\sigma \Leftrightarrow dQ = f(x,y)\,d\sigma \quad (x,y)\in D$

令 $x=r\cos\theta$,　$y=r\sin\theta$




$\Delta\sigma = \frac{1}{2}(r+\Delta r)^2\Delta\theta - \frac{1}{2}r^2\Delta\theta = \underbrace{r\Delta\theta\Delta r}_{\text{线性主部}} + \underbrace{\frac{1}{2}\Delta r(\Delta\theta\Delta r)}_{o(\Delta\theta\Delta r)}$

$\therefore d\sigma = r\Delta\theta\Delta r = r\,d\theta\,dr$

得 $dQ = f(r\cos\theta, r\sin\theta)\,r\,d\theta\,dr$

注意：将直角坐标系的二重积分化为极坐标系下的二重积分需要进行“三换”：

1. 坐标变换：$\begin{cases} x = r\cos\theta \\ y = r\sin\theta \end{cases}$

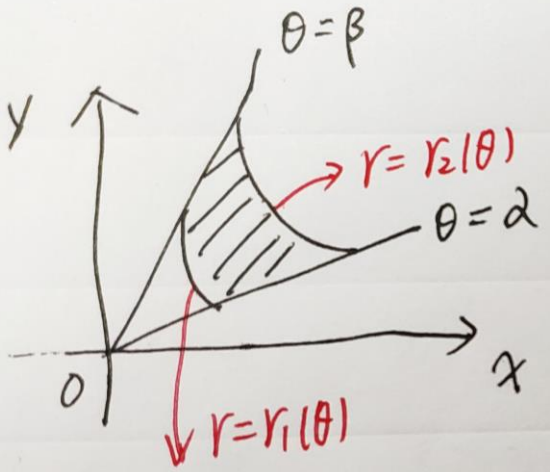
2. 微元变换：$d\sigma = dxdy = rdrd\theta$

3. 区域变换：$D_{xy} \to D_{r\theta}$

$$\iint_D f(x,y)dxdy = \iint_D f(r\cos\theta, r\sin\theta)rdrd\theta.$$

**θ-型区域**：设D是平面有界闭区域，从极点O出发的任何一条射线与D的边界至多有两个交点（D的边界是射线的一段除外），称D为θ-型区域且

$$D:\begin{cases} r_1(\theta)\le r\le r_2(\theta) \\ \alpha\le\theta\le\beta \end{cases}$$




当$\theta\in[\alpha,\beta]$时，$r_2(\theta)\ge r_1(\theta)$

则 $\iint_D f(x,y)\,d\sigma$

$$=\iint_{D_{r\theta}} f(r\cos\theta, r\sin\theta)\,r\,d\theta\,dr$$

$$=\int_\alpha^\beta d\theta\int_{r_1(\theta)}^{r_2(\theta)} f(r\cos\theta, r\sin\theta)\,r\,dr$$

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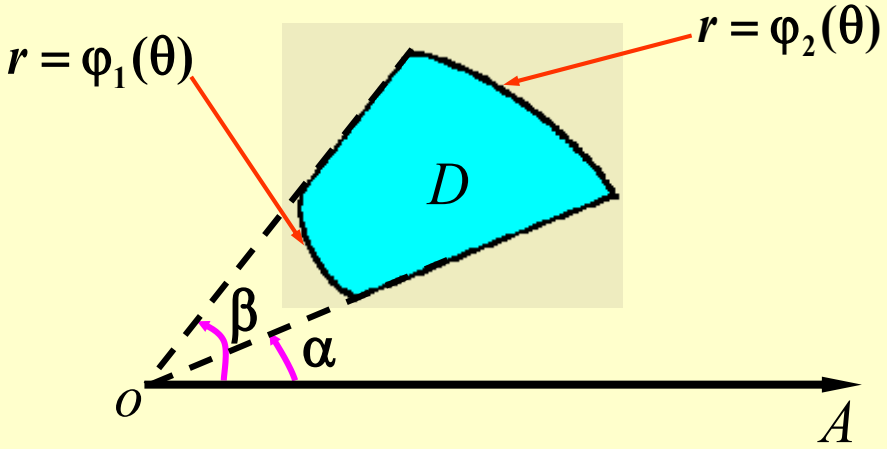
# 将二重积分化为累次积分的公式

## 1. 极点在区域的外面

区域特征如图

### $\theta$—型区域

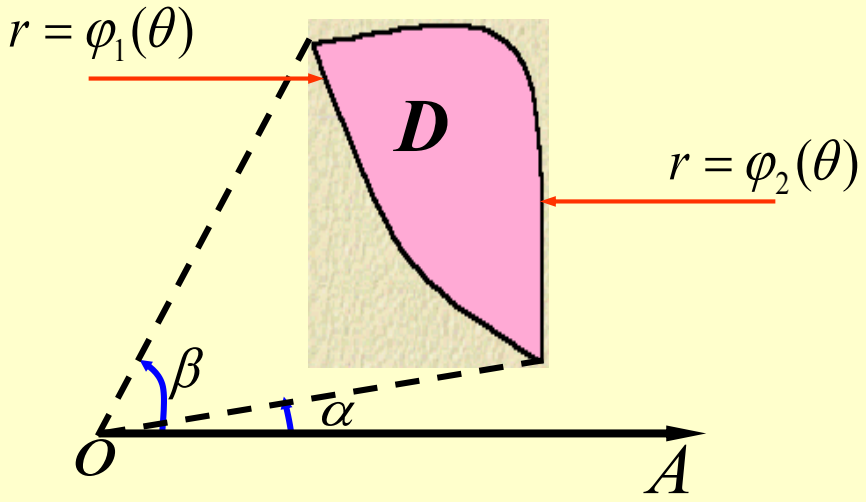
$$D:\begin{cases}\varphi_1(\theta)\le r\le\varphi_2(\theta)\\ \alpha\le\theta\le\beta\end{cases}$$




$$\iint_D f(r\cos\theta,r\sin\theta)r\,dr\,d\theta$$

$$=\int_\alpha^\beta d\theta\int_{\varphi_1(\theta)}^{\varphi_2(\theta)} f(r\cos\theta,r\sin\theta)r\,dr.$$

区域特征如图



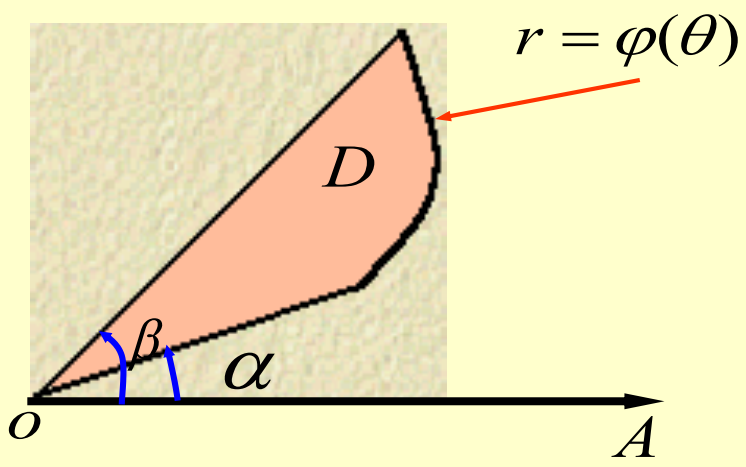

$$D:\begin{cases}\alpha\le\theta\le\beta\\ \varphi_1(\theta)\le r\le\varphi_2(\theta)\end{cases}$$

$$\iint_D f(r\cos\theta,r\sin\theta)r\,dr\,d\theta$$

$$=\int_\alpha^\beta d\theta\int_{\varphi_1(\theta)}^{\varphi_2(\theta)} f(r\cos\theta,r\sin\theta)r\,dr.$$

## 2. 极点在区域的边界上

区域特征如图




$$0 \le r \le \varphi(\theta),$$

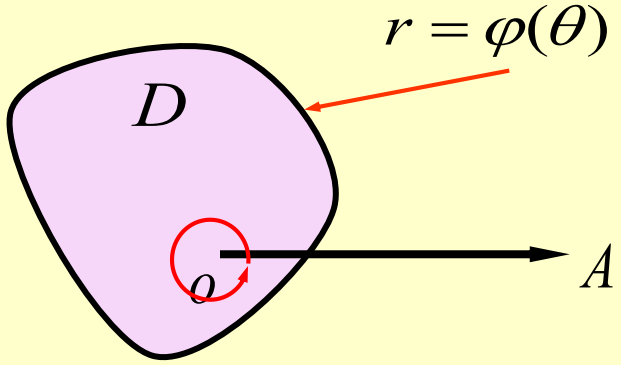
$$\alpha \le \theta \le \beta.$$

$$\iint_D f(r\cos\theta, r\sin\theta) r\,dr\,d\theta$$

$$= \int_\alpha^\beta d\theta \int_0^{\varphi(\theta)} f(r\cos\theta, r\sin\theta) r\,dr.$$

## 3. 极点在区域的内部

区域特征如图




$$0 \le r \le \varphi(\theta), \quad 0 \le \theta \le 2\pi.$$

$$\iint_D f(r\cos\theta, r\sin\theta) r\,dr\,d\theta$$

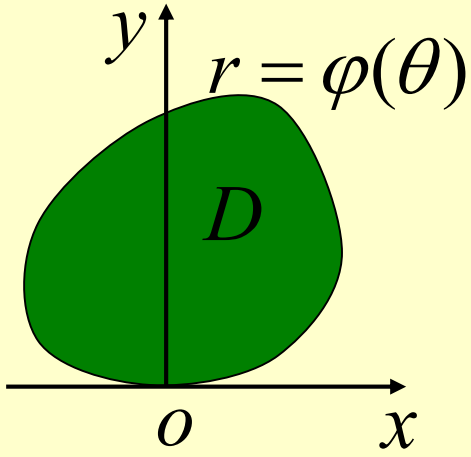
$$= \int_0^{2\pi} d\theta \int_0^{\varphi(\theta)} f(r\cos\theta, r\sin\theta) r\,dr.$$

若 $f \equiv 1$ 则可求得 $D$ 的面积

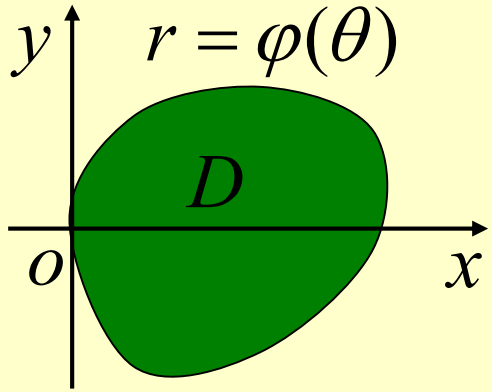
$$\sigma = \iint_D d\sigma = \iint_D r\,dr\,d\theta = \frac{1}{2}\int_0^{2\pi} \varphi^2(\theta)\,\mathrm{d}\theta$$

思考: 下列各图中域 $D$ 分别与 $x$ , $y$ 轴相切于原点,试问 $\theta$ 的变化范围是什么?

(1)




(2)

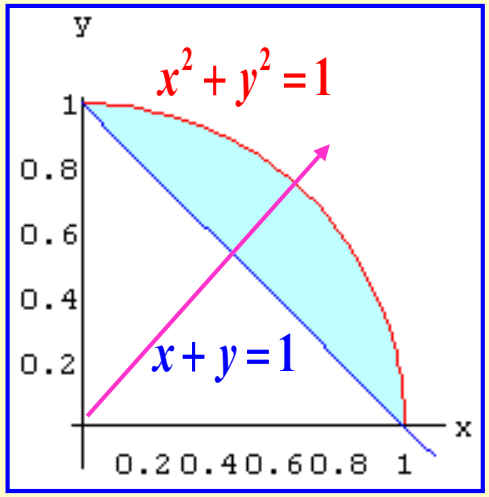



答: (1) $0 \le \theta \le \pi$; (2) $-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$

**例 1** 写出积分$\iint_D f(x,y)dxdy$的极坐标二次积分形式，其中积分区域

$$D=\{(x,y)\mid 1-x\le y\le\sqrt{1-x^2},\ 0\le x\le 1\}.$$

**解** 在极坐标系下$\begin{cases}x=r\cos\theta\\ y=r\sin\theta\end{cases}$

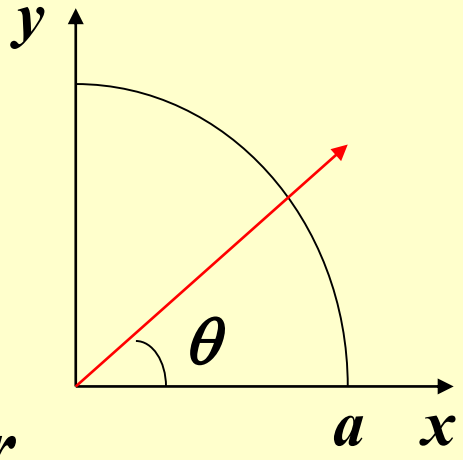



所以圆方程为$r=1$,

直线方程为$r=\dfrac{1}{\sin\theta+\cos\theta}$,

$$\iint_D f(x,y)dxdy=\int_0^{\frac{\pi}{2}}d\theta\int_{\frac{1}{\sin\theta+\cos\theta}}^{1}f(r\cos\theta,r\sin\theta)rdr.$$

例 2　计算$\iint_D e^{-x^2-y^2}dxdy$，其中 $D$ 是由中心在原点，半径为 $a$ 的圆周所围成的闭区域.



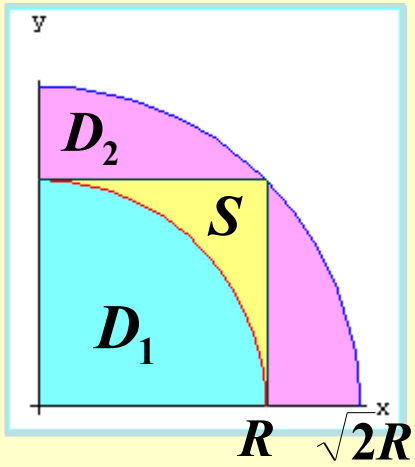

解　在极坐标系下

D：$0 \le r \le a$，$0 \le \theta \le 2\pi$

$$\iint_D e^{-x^2-y^2}dxdy = \int_0^{2\pi} d\theta \int_0^a e^{-r^2} r dr$$

$$= \pi(1-e^{-a^2}).$$

例 3　求广义积分 $\int_0^{\infty} e^{-x^2}dx$.




解　$D_1=\{(x,y)\,|\,x^2+y^2\le R^2\}$

$D_2=\{(x,y)\,|\,x^2+y^2\le 2R^2\}$

$S=\{(x,y)\,|\,0\le x\le R, 0\le y\le R\}$

$\{x\ge 0, y\ge 0\}$　显然有 $D_1\subset S\subset D_2$

$\because\quad e^{-x^2-y^2}>0,$

$$\therefore\quad \iint_{D_1} e^{-x^2-y^2}dxdy<\iint_{S} e^{-x^2-y^2}dxdy<\iint_{D_2} e^{-x^2-y^2}dxdy.$$

又$\because$ $$I=\iint_{S} e^{-x^2-y^2}dxdy=\int_0^R e^{-x^2}dx\int_0^R e^{-y^2}dy=\left(\int_0^R e^{-x^2}dx\right)^2$$

由上题结论

$$I_1=\iint_{D_1} e^{-x^2-y^2}dxdy=\int_0^{\frac{\pi}{2}}d\theta\int_0^R e^{-r^2}rdr=\frac{\pi}{4}(1-e^{-R^2})$$

$$I_2=\iint_{D_2} e^{-x^2-y^2}dxdy=\frac{\pi}{4}(1-e^{-2R^2});$$

$$\because I_1<I<I_2,$$

$$\therefore \frac{\pi}{4}(1-e^{-R^2})<(\int_0^R e^{-x^2}dx)^2<\frac{\pi}{4}(1-e^{-2R^2});$$

当$R\to\infty$时，$I_1\to\frac{\pi}{4}$，$I_2\to\frac{\pi}{4}$，

故当$R\to\infty$时，$(\int_0^{\infty} e^{-x^2}dx)^2=\frac{\pi}{4}$ 即 $\int_0^{\infty} e^{-x^2}dx=\frac{\sqrt{\pi}}{2}$